PES UNIVERSITY

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Department of Computer Science and Engineering

UE24MA241B – Linear Algebra and Its Applications

MINI PROJECT

Students are encouraged to explore the applications of Linear Algebra through the development of a mini project. This project is designed to help you understand how mathematical concepts are used to solve real-world problems by transforming raw data into meaningful results.

The following workflow template will guide the design and development of your Linear Algebra mini project.

Guidelines for Mini project

The diagram shows a **step-by-step pipeline** that begins with real-world data and ends with a useful application outcome.

Step 1: Start with Real-World Data

Choose a dataset related to a practical problem.
Examples: ratings, signals, images, traffic data, population data, etc.

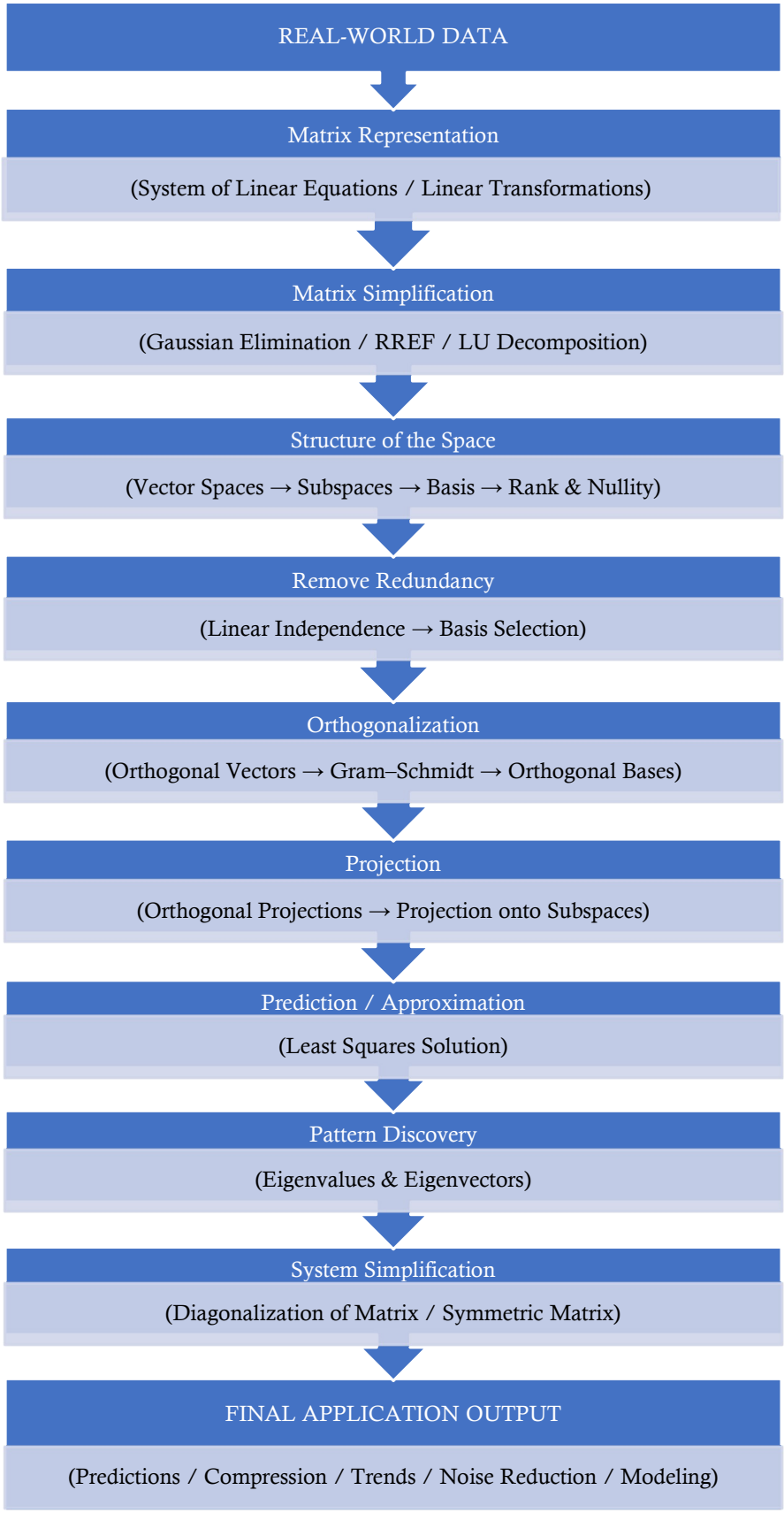
Your data must be something that can be represented in **matrix form**.

Step 2: Choose whichever Intermediate Stage applies

From the workflow diagram, select whichever stage applies between the first and last steps.

Step 3: End with a Final Application Output

Your project must end with a meaningful outcome such as Predictions (e.g., missing values, future trends) / Pattern detection/ Data compression/ Noise reduction/ Modeling or simulation


REAL-WORLD DATA
Matrix Representation
(System of Linear Equations / Linear Transformations)
Matrix Simplification
(Gaussian Elimination / RREF / LU Decomposition)
Structure of the Space
(Vector Spaces → Subspaces → Basis → Rank & Nullity)
Remove Redundancy
(Linear Independence → Basis Selection)
Orthogonalization
(Orthogonal Vectors → Gram–Schmidt → Orthogonal Bases)
Projection
(Orthogonal Projections → Projection onto Subspaces)
Prediction / Approximation
(Least Squares Solution)
Pattern Discovery
(Eigenvalues & Eigenvectors)
System Simplification
(Diagonalization of Matrix / Symmetric Matrix)
FINAL APPLICATION OUTPUT
(Predictions / Compression / Trends / Noise Reduction / Modeling)

Mini Project Evaluation Scheme

Demo Evaluation – 5 Marks

Your demo will be assessed based on:

1. **Execution**
    - Successful running of the program
2. **Concepts Covered**
    - Correct use of selected linear algebra components from the workflow
3. **Output of Each Component**
   You must clearly explain the result produced at each stage, such as:
    - Matrix representation of data
    - RREF / matrix simplification
    - Basis and orthogonal basis formation
    - Projection-based prediction
    - Least squares estimation
    - Eigenvalue/eigenvector analysis
    - Final reduced model or application output

Viva Evaluation – 5 Marks

Your viva will test **understanding, not memorization**.

For **each linear algebra concept** used in your project, you must explain:

- **What concept was used**
- **Why it was needed in the project**
- **What result it produced on the data**

How to Answer in Viva

Follow this simple structure:

Concept → Purpose → Outcome

What Examiners Will Focus On

- What your data matrix represents
- Why each mathematical step was applied
- How each step improved prediction or revealed patterns
- How one step connects to the next in your workflow

What You Are NOT Expected to Do

- Long derivations
- Theorem proofs
- Heavy manual calculations